Supplemental Notes to "Dark Matter Behaves Like Matter: A Halo Rule Set by Starlight Predicts Dwarf Galaxies Better Than MOND and Two Modified Gravity Theories"

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About these notes

These twelve notes give in full the tests that the paper summarizes: each sensitivity check, audit, failed attempt and forecast, in the wording of the paper's long version of 10 November 2026. Each note is headed by the section of the paper that it supports. Section and table numbers without an S refer to the paper. The protocols and forecasts named here, with their hashes, and the scripts that reproduce every number are in the public repository named in the paper's Data availability statement: https://github.com/MatheismOrg/papers, folder SUPPLEMENTS/A19_Reproduce.

S1. Dwarfs: second compilation, out-of-sample test, learning curve, external field and environment (Section 3.3)

The transfer holds on a second compilation. On a fixed version of the Local Volume Database (Pace, 2025), which gives each measurement with its source, 42 dwarf spheroidals with measured dispersions, 33 of them among the 44, give the same ranking. With the same circularized radii and frozen rules, the stellar rule scores 0.389 dex against 0.611 for the radial acceleration relation and 0.612 for the baryonic-mass halo, 36 percent lower than the relation; with major-axis radii, 0.367 against 0.613 and 0.616.

A test on dwarfs outside the original sample answers the order of choices. Under a written protocol, with the rules frozen on the disks, the 25 dwarf spheroidals in the same database that are not among the 44 and pass the same selection were scored; 17 of them were scored blind, under a protocol fixed and hashed before any was seen. The stellar rule scores 0.441 dex against 0.808 for the radial acceleration relation and 0.802 for the baryonic-mass halo, 45 percent lower; the paired bootstrap interval of its gain over the relation, -0.449 to -0.272 dex, excludes zero, and 21 of the 25 dwarfs improve. On the 17 scored blind the figures are 0.405 against 0.776. The ranking holds for major-axis radii and mass-to-light ratios of 1 and 3. The published Moster relation, a second halo set by starlight and fitted to no rotation data, does slightly better on these dwarfs, 0.403 dex (0.397 on the 17 scored blind), and worse on the original 44 and the second compilation, 0.556 and 0.463 dex. Both starlight rules beat MOND's relation on all three dwarf sets.

A learning curve shows the transfer is learned from the disks, not tuned to any of them. Holding out a random 5 to 90 percent of the disks, 50 times at each fraction, and refitting every rule on the rest, the stellar rule beats the radial acceleration relation on both dwarf sets in every draw from 30 training disks up, and in at least 39 of 50 draws even from 15. As training grows, its dwarf errors settle, from 0.500 to 0.478 dex on the original 44 and near 0.44 on the 25 out of sample, their spread across draws shrinks five- to sevenfold, and its three coefficients converge; on the held-out disks the three rules stay tied.

MOND's defenders answer that a satellite's own gravity is weakened by its host's field, the external field effect, which Chae et al. (2021) report in the rotation curves of disk galaxies. Applied as McGaugh and Milgrom (2013) apply it to Andromeda's dwarfs, with the host's field V^2/D, V = 230 km/s for Andromeda and 200 km/s for the Milky Way, it lowers MOND's predictions for dwarfs that already

move faster than MOND predicts: its error rises to 0.96 dex on the original 44 and 1.20 on the 25 out of sample, and stays between 0.91 and 1.00 on the 44 for Milky Way speeds of 175 to 220 km/s. On the 25 brighter dwarfs of the original sample it matches the stellar rule, 0.41 against 0.40 dex; on the 19 ultrafaint ones it misses by 1.39 dex. MOND's answer for those, that they are out of equilibrium, needs evidence that one point per dwarf cannot give.

The halo has an environment too. In simulations, a subhalo's concentration rises toward its host's center by a factor 1 - 0.54 log10(R/R_200) (Moliné et al., 2017). The factor describes concentration at a satellite's present distance, not stripping along its orbit. Applied to the stellar rule as a radial factor on its baseline concentration, with that published coefficient and host masses of 1.0 x 10^12 and 1.5 x 10^12 suns, it lowers the error to 0.457 dex on the original 44 and 0.383 on the 25 out of sample, below the Moster relation on both (0.556 and 0.403), mainly by lifting the ultrafaint dwarfs nearest their hosts. It was tried after the residuals were seen to grow toward the host, so it was kept out of the rule as a lead for the next blind test, which rejected it (Section 3.4). In an exploratory check with measured pericenters (Pace, Erkal and Li, 2022), the stellar rule's residuals for 31 Milky Way satellites show no trend with pericenter or present distance at fixed luminosity (rank partial correlations 0.05 and -0.11), while at fixed pericenter fainter dwarfs move faster than predicted (-0.41, p = 0.02). The faint dwarfs are the nearby ones (rank correlation of distance with luminosity 0.50), so the trend toward the host is a trend with luminosity. Binary stars could be part of it: in the same multi-epoch data, modeling binaries and instrumental offsets lowers Reticulum II's dispersion from 3.5 to 2.8 km/s (Minor et al., 2019), and the two binaries found in Boötes II raise its dispersion by about 25 percent, to 3.6 from 2.9 km/s (Bruce et al., 2023).

S2. Dwarfs no rule had scored (Section 3.4)

The database holds two kinds of dwarfs that no earlier test scored. Some lie beyond the Milky Way and Andromeda systems, and some have only an upper limit on their dispersion. Dropping the limits drops the slowest dwarfs, which favors any rule that predicts fast motion. A third protocol, written and hashed before any prediction for these dwarfs was computed, scored both kinds with every rule frozen (Table S1).

Table S1. Dwarfs no rule had scored, every rule frozen: error on the five with measured dispersions, in dex, and upper limits exceeded, of ten.

Model	Five measured, dex	Limits exceeded
Halo set by stellar mass and size	0.171	2
Stellar rule with subhalo factor	0.171	3
Radial acceleration relation, disk a0	0.202	0
Radial acceleration relation, published a0	0.185	0
MOND with external field	0.208 (four dwarfs)	0
Halo set by baryonic mass	0.225	0
Published Moster relation	0.235	4

Five dwarfs have measured dispersions: Cetus, Phoenix, Tucana and UGC 4879, which have no host galaxy, and KDG 64 in the M81 group. The first four are far from any large galaxy, so they feel almost no tides or external field, and neither of MOND's answers for satellites applies to them. The stellar rule has the lowest error of seven models, 0.171 dex against 0.202 for MOND's relation, 0.208 for MOND with the external field on the four it covers, and 0.235 for the Moster relation. Five dwarfs cannot make that significant: the bootstrap interval of the gain over MOND's relation runs from -0.170 to +0.144 dex, so the protocol's first test is not met. The ranking is also fragile. With major-axis radii, MOND's relation with its published a0 scores lower than the stellar rule, 0.186 against 0.201 dex,

and at a mass-to-light ratio of 3 every version of MOND does.

Ten satellites have only upper limits, at 95 percent confidence in the database. The stellar rule's predicted dispersion lies below eight of them. It exceeds the limit of Grus II, 3.3 against 2.0 km/s (Simon et al., 2020), and of Andromeda XII, 5.2 against 4.0 km/s (Collins et al., 2013); only Grus II lies beyond the rule's own scatter, a factor of 1.46 in dispersion. MOND meets all ten, so the protocol's second test is not met either. MOND passes because it predicts slow motion for every faint dwarf. The same predictions miss the faint dwarfs with measured dispersions in Section 3.3, 19 in the original sample and 17 of the 25 outside it, by 1.39 and 1.35 dex in root-mean-square, factors above 20 in acceleration. The Moster relation, which predicts faster motion than the stellar rule, exceeds four limits. These counts hold for major-axis radii, mass-to-light ratios of 1 and 3, and Milky Way speeds of 175 and 220 km/s.

Upper limits can catch only a rule that predicts too much motion. Measured dispersions catch errors both ways, and on all four dwarf sets with measured dispersions the stellar rule is ahead of MOND's relation. Section 3.5 weighs both kinds in one score. The subhalo factor of Section 3.3 raises the predicted dispersions of satellites close to their hosts, where several limits are lowest, and exceeds three limits, one more than the rule without it. The lead fails its test and stays out of the rule.

S3. All 85 dwarfs in one score: sensitivity and held-out checks (Section 3.5)

The gain over MOND's relation stays between 27 and 48 units for mass-to-light ratios of 1 and 3, for major-axis radii, without measurement errors, and with every measured dispersion below 3 km/s dropped. The last checks binary stars, which inflate the smallest dispersions most; modeling the binaries themselves needs repeated spectra of each star and is left for later work. The Moster relation trails

the stellar rule by 15 units (3 to 25). The subhalo factor scores 2.4 units better than the rule without it, inside its bootstrap interval of -8.6 to +3.3, so its net effect is not established and it stays out of the rule.

Two checks added after the run score each dwarf held out in turn. With each model's scatter fitted to the other 84 dwarfs, the stellar rule leads MOND's relation by 35.7 units, with a bootstrap interval over dwarfs of 21.8 to 47.8, and 70 of the 85 dwarfs favor it. When both may also shift their predictions, again fitted without the dwarf scored, the lead is 8.8 units (1.4 to 15.4). With that shift the baryonic-mass halo comes within 4.2 units (-4.4 to 11.4): its loss on dwarfs is mostly one uniform shortfall of 0.45 dex, and setting that level right from the disks alone is what the stellar rule adds. These scores share the likelihood's simplifications: an upper limit counts as an exact ceiling, the errors that size and distance share between predicted and measured accelerations are left out, and the bootstrap treats satellites of one host as independent.

S4. What does not help (Section 3.7)

Published feedback profiles, which change halos by the gas that star formation drives out and which fit individual galaxies well (Li et al., 2020), do not improve the shared rule: the DC14 recipe (Di Cintio et al., 2014) gives 0.189 dex on disks, a mixture with NFW chooses a 5 percent share and gains nothing, and coreNFW (Read, Agertz and Collins, 2016) with an assumed formation time gives 0.163 dex. These cored profiles are the usual answer to the cusp-core problem, and the shared rule does better without them. That holds on the benchmark's points and for cores given to every galaxy by a published recipe; on the noisier points that the benchmark's quality cut removes, a core confined to diffuse gas-rich disks is needed, and the final formula carries one (Sections 6.1 and 6.4). A blend of the radial acceleration relation and the stellar halo, weighted per training split, lowers

the disk error by under 2 percent, a gain its uncertainty does not establish, and transfers to dwarfs less well than the stellar halo alone (0.505 dex). Allowing halo scatter improves a predictive log score by 0.19 units per radius, with a 95 percent interval of 0.09 to 0.30, without improving the central error; the stellar rule's gain is in the point prediction itself.

S5. Whole halos from lensing (Section 3.8)

Rotation curves reach a few tens of kpc, while galaxy-galaxy lensing weighs whole halos. Mandelbaum et al. (2016) measured the mean halo masses of red and blue central galaxies in the Sloan survey, in bins of stellar mass. Under an eighth protocol, hashed before any prediction, the stellar rule frozen on the disks, given the disks' own scalings of size and gas with stellar mass, predicts the halos of the blue, star-forming centrals within 0.16 dex in four of five bins from 10^10.2 to 10^11.3 suns, a median 0.05 dex high. At 10^10.56 suns it is 0.34 dex high, where the measured mass dips below that of the bin beneath, so the protocol's test, every bin within 0.2 dex or twice its error, is not met. Means over a bin exceed typical masses by roughly 0.05 to 0.1 dex, which the comparison leaves uncorrected. For red centrals half the size of disks, the rule matches the measured halos up to 10^10.9 suns but gives only about half of the measured red-blue split there, 0.17 to 0.20 dex against 0.35 to 0.41. Above 10^11 suns the red centrals' halos grow about as the square of stellar mass and lie above the rule by 0.3 dex at 10^11.1 suns and 1.0 dex at 10^11.7: the bend of Section 3.7, reached by the centrals of groups and clusters. X-ray masses of 21 clusters show the same steepness from the other side: their central galaxies' stellar mass grows only about as M_500^0.4, with about 0.2 dex scatter (Kravtsov, Vikhlinin and Meshcheryakov, 2018). The inverted Moster relation lies 0.4 to 1.0 dex above the red centrals' mean halos over the same range.

Lensing also traces how the mass is spread. Brouwer et al. (2021) stacked the Kilo-Degree Survey (KiDS-1000) lensing signal of 259,383 isolated galaxies at redshifts 0.1 to 0.5 in 15 bins of baryonic acceleration, spanning about 30 kpc to 3 Mpc, for four bins of stellar mass and for early and late types of matched stellar mass. A ninth protocol, hashed before any measured value was read, as were the predicted profiles, carries the frozen rule into these stacks. It uses the published stellar masses, the disks' scalings of size and gas for late types, and, for early types, no gas and the median size gap between early and late types at each stellar mass in Hubble imaging (van der Wel et al., 2014). Inside 0.3 Mpc, where false isolated lenses add little, the measured signal lies 0.027 ± 0.015 dex below the prediction for the whole sample, between 0.005 and 0.048 dex below it in each of the four stellar-mass bins from 10^8.5 to 10^11 suns, and 0.00 and 0.01 dex below it for red and bulge-dominated galaxies. For blue and disk-dominated galaxies it lies 0.14 and 0.12 dex below, outside the protocol's 0.1 dex; the rule likewise exceeds the halo masses of Mandelbaum's blue centrals of 10^10.5 to 10^11 suns, by 0.13 to 0.34 dex. At redshift 0.3, near KiDS's 0.25, CFHTLenS halo-model masses of blue galaxies, with satellites modeled, instead lie within 0.07 dex of the rule at 1.7 and 4.2 × 10^10 suns (Hudson et al., 2015), so the surveys do not yet agree for these galaxies. The rule therefore puts early types only 0.05 dex above late types of the same stellar mass, against a measured 0.16 dex by Sérsic index and 0.21 dex by color inside 0.3 Mpc: about a third and a quarter of the split, which the protocol counts as part of the first and not the second. Beyond 0.3 Mpc the whole sample's signal exceeds the single-halo model by 0.16 to 0.57 dex, where Brouwer et al. expect neighbors missed by the isolation criterion to add signal. An independent script reproduces every amplitude to 0.0001 dex.

These are amplitudes; the radial profile does not match. With the release's covariance, the whole sample's seven bins inside 0.3 Mpc give a chi-square of 48.6 against the frozen prediction and 45.1 for six degrees of freedom once the amplitude is fitted (p = 4 × 10^-8). The measured signal lies 0.19 dex below the prediction in the innermost bin and 0.16 dex above it in the bin nearest 0.3 Mpc, so the measured profile is flatter than the predicted one. Three of the four stellar-mass bins and the blue and bulge-dominated galaxies also miss the profile, at p = 10^-4 to 0.03, while the red and disk-dominated galaxies match it (chi-square 7.6 and 8.1 for six). The protocol's lensing model extends each halo beyond r_200, where the rotation-curve fits truncate it; halos truncated at r_200 move the amplitudes by 0.003 dex or less and still miss the profile (40.6 for six), as do all 14 of the protocol's sensitivity cases (39.5 to 50.1). These tests take the model as exact and use only the measurement covariance, and since the samples overlap, their p-values are not independent.

Lensing does not yet support the rule's size term at fixed stellar mass and type. In KiDS near 3 × 10^10 suns, halo mass follows size and Sérsic index more closely than color, with a suggestion that compact early types sit in heavier halos while late types share about one halo mass (Taylor et al., 2020). In CFHTLenS, at fixed stellar mass and color, halo mass grows with size, as size to the power 0.28 ± 0.18 for blue galaxies and 0.53 ± 0.15 for red ones, and 0.22 ± 0.42 for blue galaxies near 10^10.6 suns, with satellites kept in the samples and concentration fixed by mass (Charlton et al., 2017); the rule's halo mass instead falls as size to the power of about -0.6 there. Charlton et al. note that with concentration fixed, part of a size trend could be a change of concentration instead. A lensing measurement of halo mass against size at fixed stellar mass and type, for central galaxies and with the rule's prediction fixed first, would decide it.

S6. A published halo relation at cluster scale, and its two audits (Section 5.3)

The halo side can be carried across the same gap, as an exploratory extension that rests on the published Moster relation alone; the paper's own three-coefficient rule remains a disk-to-dwarf result. The Moster relation of Sections 3.2 to 3.5, unchanged, maps a central galaxy's stellar mass to its host halo's mass up to about 10^15 suns. A fifth protocol, hashed before any prediction was computed, applies it with nothing fitted to the 50 MaNGA brightest cluster galaxies and the 20 CLASH clusters of Section 2.1. Each brightest galaxy's published stellar mass and redshift set its halo, the concentration follows Dutton and Macciò (2014) at that redshift, and the measured baryons are added (Table 6); the CLASH stellar masses are those of Cooke et al. (2016), which Tian et al. take as total. Moster, Naab and White (2013) define halo mass at 200 times the critical density, as the concentration relation does, and give masses in suns for h = 0.704; as in Section 3.1, the halo's dark mass is (1 - f_b) times the halo mass, and its pull adds to the measured baryonic pull. Each galaxy or cluster weighs equally.

Inverted at its mean, the Moster relation cuts the error of MOND's relation by more than half, 0.185 against 0.403 dex, and the bootstrap interval of the gain, -0.248 to -0.188 dex, meets the protocol's test. Its median acceleration error is 26 to 33 percent in each group, against 50 to 72 percent for MOND's relation, which falls short in every group: by 0.30 dex in the MaNGA galaxies, 0.45 dex at CLASH centers and 0.54 dex at 100 to 600 kpc. The Moster relation's largest offset is at the MaNGA centers, 0.15 dex short. The disk-calibrated stellar rule fails at 100 to 600 kpc (0.79 dex), as Section 3.7 expects, since it has no bend above the Milky Way's mass. The gain holds with the CLASH stellar masses doubled, with the MaNGA masses lowered by 0.3 dex, and without redshift evolution: 0.20 to 0.24

dex against 0.40.

The relation is nearly flat at the top, so a small change in stellar mass moves the halo mass a lot. Five of the 20 CLASH galaxies map to halos above 10^16 suns, beyond any known cluster, and these five are overpredicted by 0.02 to 0.46 dex. For two of them, MACS J0429.6-0253 and MACS J0647.7+7015, the search found no root and stopped at its upper bound, 10^16.5 suns. Without redshift evolution only one halo lies above 10^16 suns, and the error falls to 0.20 dex.

A sixth protocol, hashed before it ran, audited this test. Inverting the mean relation ignores its published scatter of 0.15 dex in stellar mass at fixed halo mass. Inverted instead with that scatter, the measurement errors and the halo mass function of Tinker et al. (2008), computed with Colossus (Diemer, 2018), the relation puts every halo's posterior median below 10^15 suns and still beats MOND's relation, 0.288 against 0.403 dex, a 29 percent cut rather than 54, with a bootstrap interval of the gain of -0.136 to -0.095 dex (Table 6). Scored only on the 50 systems whose stellar masses lie within the range the relation was fitted on, at most 10^11.7 suns, the mean inversion gives 0.187 against 0.377 dex, and with h = 0.704, as Moster's parameters assume, 0.182 against 0.403. The accelerations survive these checks; the halo masses do not.

Tian et al. computed the CLASH lensing accelerations from the spherical NFW fits of Umetsu et al. (2016), and a seventh protocol, hashed before it ran, compares the Moster masses with those published masses, M_200 of 10^14.8 to 10^15.5 suns; NFW fits to the accelerations themselves recover the published masses within 0.05 dex. The mean inversion's masses scatter by 0.73 dex about the published ones, a factor of five, where the published errors allow 0.13 dex, around a median 0.21 dex high; only 5 of the 20 lie within twice the published error. No central galaxy could do much better: with its stellar mass growing as about M_500^0.4 and scattering by about 0.2 dex (Kravtsov, Vikhlinin and

Meshcheryakov, 2018), it fixes the halo mass only to about 0.5 dex, while the cluster's total stellar mass, growing as M_500^(0.6 ± 0.1) with 0.1 dex scatter, would fix it to about 0.2 dex. The probabilistic masses fall a median 0.98 dex low, and none lies within twice the error, as expected when the mass function's prior ignores how CLASH chose its clusters: 16 for X-ray temperatures above 5 keV and smooth, symmetric X-ray images, and 4 for their high lensing magnification (Umetsu et al., 2016). The four chosen for lensing fall furthest, 1.43 dex low against 0.89 for the sixteen. With halos that light, the probabilistic predictions fall short in every group, by 0.18 to 0.40 dex. Lensing puts the mean halos of red central galaxies 0.4 to 1.0 dex below the inverted relation above 10^11.1 suns (Section 3.8), so part of the mean inversion's lead over the probabilistic one may come from that overshoot offsetting CLASH's selection of massive clusters. At these radii a halo's pull changes by only about 0.2 to 0.7 dex per dex of halo mass, and the baryonic pull not at all, so the test shows that halos set by starlight predict cluster-scale accelerations better than MOND's relation, not that the relation predicts each cluster's mass. The same published relation is within 12 percent of MOND's relation on disks and ahead of it on the dwarfs taken together, while MOND's relation needs its constant raised about six times between disks and clusters. Section 6.1 puts CLASH's selection into the prior.

S7. The one formula, step by step (Section 6.2)

The rows named in this note are those of Table 7.

Against the rule alone, the final formula (last row) keeps the disks, lowers the lensing chi-square from 79.0 to 64.5, and cuts the errors by 0.67 dex for the clusters at 100 to 600 kpc, 0.28 dex at their centers and 0.21 dex for MaNGA, with host-bootstrap intervals of the gain of -0.71 to -0.64, -0.35 to -0.22 and -0.23 to -0.18 dex. The subhalo factor and the floor improve the dwarfs by 0.036 dex

(-0.074 to -0.000). The IMF trend, with nothing refitted to MaNGA, removes the 0.15 dex by which the MaNGA galaxies sat above the formula and lowers their error from 0.184 to 0.114 dex (-0.088 to -0.050). For the early types it lowers the error from 0.203 to 0.185 dex, but their inner points come from the dynamical models that helped set the trend, and their outer, neutral-hydrogen points do not improve. Halo contraction (Cautun et al., 2020) at the strength the fit prefers adds little once the trend is in (MaNGA 0.114 to 0.111), so the formula leaves it out.

The expected halo with each sample's selection lowers the cluster errors from 0.209 to 0.136 dex at 100 to 600 kpc and from 0.155 to 0.105 at the centers (-0.120 to -0.020 and -0.099 to -0.001 dex, and the same on a stricter lensing mask without each stack's outermost bin), with MaNGA within 5 percent. Section 5.3's probabilistic inversion fell short because its prior left out how CLASH chose its clusters. The same expectation repairs the mean inversion for galaxies chosen by their light: for the red SDSS centrals of Mandelbaum et al. (2016), which the mean inversion overshoots by 0.46, 0.74 and 0.97 dex at 10^11.3, 10^11.5 and 10^11.7 suns, the formula of the fourth row, with no selection, comes within 0.16, 0.23 and 0.18 dex, and within 0.14 dex rms over the seven red bins against 0.50. At fixed stellar mass, cluster centrals sit in the heavy tail of the halo distribution, and galaxies chosen by their light near its mean. The cluster gain holds for thresholds from 3 to 7 × 10^14 suns and scatter from 0.15 to 0.20 dex; for MaNGA, a threshold 0.2 dex lower raises the error to 0.123 to 0.125 dex. With the threshold fitted without the clusters, MaNGA kept, the clusters come out at 0.139 dex at 100 to 600 kpc and 0.105 at their centers, against 0.136 and 0.105 when trained on (0.150 against 0.142 at 100 to 600 kpc on the stricter mask), and in the last five rows of Table 7 they do not depend on the threshold at all.

A blind test is less favorable to the brightest galaxy's role. A twelfth protocol, hashed before any value was read, scored the formula of the fourth row on the

216 HeCS-omnibus clusters with a measured brightest-galaxy stellar mass (Sohn et al., 2020), none used here, with the selection their authors describe, dynamical masses above about 10^14 suns. Against the caustic masses M_200, the median offset is -0.11 dex, but the rms is 0.58 dex, larger than the 0.41 dex spread of the masses about their median; without the selection the rms is 0.80 dex, and with the mean inversion 0.95. Most of the miss comes from 22 brightest galaxies below 10^11.2 suns, whose switch was largely off and which fall 1.0 dex short; above 10^11.4 suns the prediction is unbiased (-0.03 dex) and as good as the median (0.38 against 0.37 dex). The brightest galaxy's stellar mass tracks its cluster's mass only weakly, with correlation 0.34 and slope 0.46 against the formula's 1.16: where stellar mass rises only as the 0.35 to 0.4 power of halo mass, 0.15 to 0.18 dex of scatter in stellar mass becomes 0.4 to 0.5 dex in halo mass. At cluster scale the formula therefore predicts the typical mass of a selected sample, not the mass of each cluster. The fifth row, with the switch on for every central of a halo-selected sample, was written after this test; it changes no benchmark score beyond tolerance and brings the HeCS rms to 0.45 dex, but HeCS can no longer test it.

The sixth row's lensing correction lowers the held-out chi-square from 69.5 to 64.5 (44.8 to 39.4 on the stricter mask) and improves all four stacks, with every other score within 0.1 percent. It was chosen after a look at all data showed that the stacks prefer lower concentrations, and its size comes from Neto et al., not from a fit.

The seventh row takes the threshold off the benchmark. Once the switch is on for every halo-selected sample, the threshold is set only by the early types and the lensing stacks, and weakly: fitted on the benchmark, it ranges from 10^10.89 suns to its upper bound over the 60 partitions. The switch stands for the heavier halos of central galaxies without cold gas, which galaxy-galaxy lensing measures

directly: Mandelbaum et al. (2016) give the halo masses of red and of blue central galaxies of the Sloan Digital Sky Survey in seven stellar-mass bins each. In each partition, after the rule is refitted on the training disks, M_t is the value that brings the formula closest to the seven red bins, and no benchmark system enters. It comes out at 10^11.33 suns, from 10^11.31 to 10^11.41 over the 60 partitions, with no bound hits, and every held-out score stays within 0.4 percent of the sixth row's (a fifteenth protocol, hashed before the run). Only the rule's three coefficients, set on disks, are then fitted to the benchmark; every other population is predicted. The red centrals are a calibration now, not a check, and the formula follows them only to 0.13 dex rms, a chi-square of 61 for seven bins: it falls 0.17 dex short at 10^11.1 suns and overshoots by 0.16 dex at 10^11.5 and 10^11.7. The higher threshold also shrinks the KiDS early-late split that the formula gives, from 0.071 to 0.051 dex.

The eighth row reads the formula's first term consistently. M_p/f_b gives every halo at least the cosmic share of dark matter for the baryons seen in it. For disks those are stars and cold gas; for the clusters the earlier rows counted the brightest galaxy's stars alone, although most of a cluster's baryons are hot gas, which the same data give. With M_p taken as all the baryons, stars and hot gas, inside each cluster's outermost measured radius, and nothing fitted, the error at 100 to 600 kpc falls from 0.132 to 0.100 dex (-0.045 to -0.020; a sixteenth protocol, hashed before the held-out run) and the average shortfall there from 0.11 to 0.07 dex; the centers move from 0.106 to 0.102 (-0.015 to +0.008), and no other score changes. The step was found on these data. It depends on how far out the gas is measured, though weakly: with each cluster's baryons extrapolated to 1.5 and 2 times that radius the error is 0.089 and 0.088 dex, and at 3 times, 0.106. The gas alone, as a halo that closes the cosmic baryon budget inside the same radius, gives 0.143 dex and the same typical mass as the expected halo,

and the HeCS clusters, without gas measurements, cannot test the step. The sum of the two terms is the formula's form, not a derivation.

The last row gives the halos of diffuse gas-rich disks their core (a nineteenth protocol, hashed before its held-out run; a seventeenth and an eighteenth, which failed, are described in Section 6.4). The step answers a comparison that the benchmark does not make. The benchmark keeps rotation-curve points with velocity errors of at most 10 percent, and on the points it drops MOND's relation beats the formula of the eighth row (Section 6.4). In each partition A, B and C are fitted as before, on the benchmark's points, and then t and κ on all points of the same training disks, the dropped ones included, each weighted by the inverse of its variance, with the velocity error and 0.131 dex, the galaxy-to-galaxy scatter of the eighth row's formula, added in quadrature. Over the partitions t runs from 0.11 to 0.18 and κ from 0.54 to 0.71, with no bound hits, and at the values from all 149 disks the core reaches half strength in 36 of SPARC's 175 galaxies and 0.70 at most. On the benchmark the core changes little. The disks move from 0.1533 to 0.1531 dex and the lensing chi-square from 64.7 to 64.5, since the lensing, measured far outside the cores, takes only the mass term; no other score changes, since dwarf spheroidals and early types hold no cold gas and the cluster samples are chosen by their halos. Its gain is on the points the benchmark drops: held out, their error falls from 0.401 to 0.344 dex, and that of all points from 0.218 to 0.193 (Section 6.4). Five coefficients are then fitted to the benchmark, all of them on disks. The core was found on these data, after the comparison of Section 6.4 was made.

Four limits remain. First, the threshold rests on massive central galaxies chosen by their light: fitted on the benchmark without any MaNGA or CLASH system, the mean-inversion formula's threshold reaches its upper bound in 4 of 60 fits, which then miss the clusters; set on the SDSS red centrals, it is stable, but the formula

follows their halo masses only to 0.13 dex. Second, the high-mass branch needs each sample's selection, stated by CLASH in X-ray temperature and by the MaNGA program as an aim, and a stated cut is not the whole selection: the expected halo of a CLASH cluster, 7.6 × 10^14 suns, lies 0.30 dex below the clusters' lensing masses in the median (Umetsu et al., 2016), so a cut at 5 keV understates how massive the chosen clusters are; with the hot gas counted the formula's halo is 1.0 × 10^15 suns, still 0.20 dex below, and its accelerations at 100 to 600 kpc fall short by 0.07 dex on average, most of their 0.10 dex error. Third, for massive star-forming galaxies the halos are too diffuse. From the Milky Way's published stars and gas (McMillan, 2017), the formula predicts its rotation curve within 7 percent of the Gaia-based curve (Eilers et al., 2019), but with a halo of 1.8 × 10^12 suns, against 1.37 × 10^12 in McMillan's fit and 1.08 × 10^12 for a contracted halo (Cautun et al., 2020); the full contraction of Cautun et al., with the rule refitted, matches the Milky Way but leaves the disks 4 percent worse and the lensing far worse. Fourth, the KiDS profiles still prefer concentrations 0.09 to 0.27 dex below the relaxed relation at fixed mass, more than the all-halo correction's 0.03 dex, and the final formula gives a quarter to a third of the measured early-late split, 0.051 against 0.16 to 0.21 dex (0.071 in the sixth row). Most of the remaining lensing chi-square sits in two places. One is the outermost bin of each stack, next to the 0.3 Mpc beyond which Brouwer et al. (2021) find that satellites missed by the isolation criterion raise the signal: without it, the chi-square falls from 64.5 to 39.4. The other is the lightest stack, whose point at 25 kpc is negative, -7 ± 12 suns per square parsec: it lies 4.1 standard errors below the formula and 1.2 below the stars and gas alone, so any halo adds to its miss, and it alone adds 16.8 to the chi-square. On the stricter mask, the other three stacks give 3.3 to 4.9 each.

S8. Against MOND and standard halos: the better cases, the noise model and the five patterns (Section 6.3)

The final formula can be set against the two standard accounts on one footing: the same systems, radii, weights and lensing covariance, with nothing fitted to the benchmark, so that their errors on all data are already held out (Table 8). MOND is the radial acceleration relation with its published constant, a0 = 1.2 × 10^-10 m/s^2 (McGaugh, Lelli and Schombert, 2016), without the external field, which only worsens its dwarfs (Section 3.5); its lensing is its phantom mass, projected, on the assumption, as in Brouwer et al. (2021), that lensing measures the same acceleration as dynamics. Standard halos are those of the published Moster relation, as the expected halo of each sample with its selection (Section 6.1), with the same concentrations, the same all-halo correction for the lensing stacks and the same floor. Each account is given its better case: both take the IMF trend of spheroids, which lowers their early-type errors from 0.207 and 0.205 dex to 0.182 and their MaNGA errors from 0.327 and 0.201 to 0.186 and 0.116, and the standard halos leave out the subhalo factor, with which their dwarf error rises from 0.440 to 0.468 dex, and take the same baryon term for the clusters as the formula, without which they score 0.106 and 0.134 dex there. The comparison was made after the formula was frozen, and nothing in it is fitted. It is a comparison on the benchmark's points; Section 6.4 extends it to the rotation-curve points that the benchmark's cut on velocity errors removes and to galaxies outside the benchmark.

In size, the formula's errors are those of the measurements on every dynamical scale, for the 310 systems of the benchmark: 149 disks, 75 dwarfs, 16 early types, 50 MaNGA galaxies and 20 CLASH clusters, with stellar masses from 5 × 10^2 to 1.5 × 10^12 suns (Table 9). With the formula's prediction taken as the truth and each system's inputs redrawn from their published errors 2,000 times,

measurement alone produces a benchmark error of 0.144 dex for the disks, from the errors of distance, inclination, luminosity and rotation velocity in SPARC and the 25 percent in stellar mass-to-light ratio and 10 percent in HI flux adopted by Lelli et al. (2017), who likewise found the scatter of the radial acceleration relation to be largely that of its errors. It produces 0.38 dex for the dwarfs, from the errors of dispersion, radius, distance and luminosity in the Local Volume Database, and 0.146, 0.112, 0.134 and 0.088 dex for the early types, MaNGA, the CLASH centers and 100 to 600 kpc, from the errors published with each acceleration. The formula's held-out errors, 0.153, 0.370, 0.185, 0.116, 0.102 and 0.100, all lie inside the central 95 percent of the draws, the early types at the upper edge. By the size of its errors, these data cannot tell the formula from a perfect one. The test is blunt, and the formula is not alone in passing it. MOND's relation passes on the disks and the early types and fails on the dwarfs, MaNGA and the clusters. Standard halos, given the better case above, pass on every dynamical scale and fail in lensing; the paired comparisons above, which the formula wins on disks and dwarfs, are the sharper test between the two. In lensing the formula's own chi-square, 64.5 for 29 points, is too high for noise. On the 24 points left when each stack's outermost bin and the one negative point are set aside (Section 6.6) it is 21.3, against 37.2 for MOND's relation and 68.7 for standard halos. The six dynamical scales and the lensing on the stricter mask, which was fixed before the formula was built, make seven tests. The formula's lowest share among them is 3.4 percent, and a perfect formula would show a share that low in at least one of seven tests one time in five. At a level of 5 percent over the seven together, 0.7 percent each, the formula passes all seven, MOND's relation fails five and standard halos fail the lensing. On the primary mask all three fail.

Patterns within the residuals test more sharply than their size, and five exceed the noise. First, within single disks the rotation curves depart from the formula by

0.066 dex where the velocity errors give 0.045 (0.043 to 0.048). Inside 0.3 half-light radii the formula is 0.04 dex too high relative to each disk's own mean, and the slope of the residuals with radius varies from disk to disk by 0.22 where the errors give 0.07. MOND's relation and standard halos leave more within disks, 0.072 and 0.080 dex, and the formula without its core left 0.073. The two published core profiles of Section 3.7, given to every galaxy with the formula's other coefficients refitted, raise the disk error to 0.183 and 0.158 dex. From disk to disk, by contrast, the formula's offsets are those of the errors (0.134 dex against 0.136, 0.114 to 0.168), with no trend against stellar mass, size, gas fraction, surface brightness, inclination or distance. Second, at fixed stellar mass the dwarfs' residuals fall with half-light radius by 0.50 dex per dex, where the redrawn samples give -0.27 to +0.28. The extremes carry the trend: without the tenth of the dwarfs most extended for their mass, Crater II, Antlia II and Andromeda XIX, XXI, XXIII and XXV among them, and the tenth most compact, the slope is -0.21. Third, the MaNGA residuals fall with baryonic acceleration (correlation -0.59 against -0.48 to +0.17) and rise with half-light radius (+0.55); most of this lies in the published inputs (Section 6.6). Fourth, the CLASH clusters at 100 to 600 kpc lie 0.07 dex above the formula on average, where noise gives 0.02 either way. Fifth, the lensing points set aside lie off the formula together: each stack's outermost bin above it, and the innermost point of the lowest-mass stack, a negative excess surface density, four standard errors below it. None of the five is fitted here.

S9. Every rotation-curve point: the excess, two failed terms, outside checks and limits (Section 6.4)

The cut removes 582 points in 139 of SPARC's 175 galaxies, most of them inside the half-light radius. On them all three accounts predict accelerations too high, and without the core the formula does worse than MOND's relation: 0.416 dex, against 0.381 for MOND's relation and 0.426 for standard halos, with average excesses of 0.21, 0.17 and 0.22 dex (formula without the core minus MOND's relation, +0.023 to +0.047). The difference lies in the 201 points that rotate slower than 30 km/s, where the three predict too much by 0.48, 0.36 and 0.49 dex on average and score 0.565, 0.471 and 0.578 dex; on the faster points they tie, at 0.323, 0.326 and 0.326. With no cut at all, over the 3,389 usable points of all 175 galaxies, they score 0.260, 0.244 and 0.274 dex. The tie with MOND's relation on disks (Section 3.2) therefore held for points measured to 10 percent: in the slowly rotating centers of small disks, cuspy halos predict too much. Pressure support does not explain the excess. On the seven galaxies shared with Iorio et al. (2017), whose circular velocities are corrected for it, SPARC's speeds at the dropped points, brought to the same inclination, agree with the corrected ones (median ratio 1.00, and 1.05 below 30 km/s).

Two terms aimed at this failed their protocols, a seventeenth and an eighteenth, each hashed before its held-out run. Weighting cold gas by a fitted factor μ in the baryon term, $M_p = M_{star} + \mu M_{gas}$, gave μ between 2.3 and 7.2, improved the dwarfs from 0.370 to 0.357 dex (-0.035 to +0.008), worsened the early types from 0.185 to 0.191 (+0.002 to +0.010) and raised the lensing chi-square from 64.7 to 75.7, outside its tolerance. Larger, looser halos for diffuse gas-rich disks, with the halo mass multiplied by $10^{(t\ g\ L)}$ and the concentration by $10^{(-s\ g\ L)}$, lowered the held-out disk error from 0.1533 to 0.1513 dex and the lensing chi-square from 64.7 to 63.8 and changed nothing else; but the protocol's mark was a disk gain

beyond 1 percent with a paired bootstrap range below zero, and the range ran from -0.005 to +0.002. Beyond the benchmark, with its two numbers frozen and each check hashed first, that term lowered the error of the 17 SPARC galaxies that the benchmark excludes for quality or inclination and that keep some point from 0.317 to 0.310 dex (-0.018 to +0.003), and of the 582 dropped points from 0.416 to 0.392 (-0.030 to -0.018), two thirds of the way to MOND's relation. On the eight dwarf irregulars outside SPARC of Table 10 its check failed as worded: with the benchmark's cut only three of them keep any point, and on those it raised the error from 0.190 to 0.197 dex (-0.020 to +0.033). With every point kept, a variant chosen after that result, it lowered the error from 0.411 to 0.357 (-0.085 to -0.021).

Two outside checks, forecast in the protocol with every coefficient frozen and every point kept, were also met. Iorio et al. (2017) give circular velocities, corrected for asymmetric drift, and HI surface densities for 17 LITTLE THINGS dwarf irregulars. Seven are in SPARC, and eight of the other ten have 3.6 micrometer fluxes in the Local Volume Legacy survey (Dale et al., 2009) and stellar scale lengths in Read et al. (2017), from which stellar masses and sizes were built on SPARC's scale, by the median ratios on the seven shared galaxies. On those eight the formula's error is 0.308 dex, against 0.411 without the core and 0.414 for MOND's relation (-0.181 to -0.032 against MOND's relation; lower in six of the eight). On the 26 SPARC galaxies outside the benchmark, excluded for quality or inclination or left without a benchmark point, it is 0.392, against 0.435 and 0.415 (-0.036 to -0.007). Against MOND's relation with the refitted constant the first check still holds (0.392; -0.154 to -0.017) and the second becomes a match (0.396; -0.016 to +0.007).

The limits of this test were fixed in the protocol. It is retrospective: the dropped points were first scored for this paper, and about a dozen core forms were looked at on all data before the protocol. The variable g L is inherited from the term that

missed its mark, and the eight dwarf irregulars had already been scored with that term. With the benchmark's cut, the three of them that keep points are fitted worse with the core, 0.196 against 0.190 dex, and 0.169 for MOND's relation. And the dropped points remain the one place where no account reaches the measurement errors. If the formula were exact, the published errors, with the cut applied to the redrawn velocities, would give 0.273 dex on the 582 points (0.235 to 0.352). With its coefficients from all 149 disks the formula gives 0.360, a result that 2 percent of such draws reach; MOND's relation and standard halos, at 0.381 and 0.426, lie further out, at 1 and 0.5 percent. In those draws the formula would beat MOND's relation by 0.015 dex (0.001 to 0.028), and it beats it by 0.021.

S10. What physics the formula stands for (Section 6.5)

The stellar term follows abundance matching for disks up to about 5 × 10^10 suns: for a disk of typical size and gas, the rule's halo lies within 0.25 dex of the expected halo of the Moster relation from 10^8 to 5 × 10^10 suns. Above that it keeps rising as a single power of stellar mass, without the break of abundance matching, so that a disk of 2 × 10^11 suns has turned a quarter of its share of baryons into stars, against 4 percent for the Moster relation. Rotation-curve fits to SPARC find the same unbroken relation, with about half (Posti, Fraternali and Marasco, 2019), and in the EAGLE simulation disks hold up to 1.5 times the stellar mass of ellipticals in halos of the same mass (Correa and Schaye, 2020). The switch is the other side of the same split: at fixed stellar mass, red central galaxies sit in halos more than twice as massive as blue ones (Mandelbaum et al., 2016), a split that EAGLE traces to differences in halo formation time and black-hole feedback (Correa and Schaye, 2020). The size term has the sign of simulated galaxy formation, in which galaxies are smaller in more concentrated halos (Jiang et al., 2019; Sun, Jiang and Wang, 2026): at fixed stellar mass a

compact disk sits in a denser halo, which the rule, with concentration tied to mass, reads as a heavier one. And the regularity that MOND's relation describes emerges in cold-dark-matter simulations, with a scatter of 0.08 dex (Ludlow et al., 2017), although EAGLE needs an acceleration scale 2.2 times higher, because its stellar masses fall short near the knee of the halo relation; the rule, set on observed disks, matches the observed relation (Table 8). The core has its counterpart too. In simulations, gas that star formation drives in and out of a dwarf's center turns its cusp into a core about the size of the stellar half-light radius, the more so the longer stars keep forming (Read, Agertz and Collins, 2016), and most where the stars are a few thousandths of the halo's mass (Di Cintio et al., 2014). The formula's cores sit where that picture puts them, in diffuse disks that still hold their gas, and are absent from dwarf spheroidals and cluster centers, which need their density high.

Three readings that go beyond cold dark matter were tested on all data, on the benchmark's points, with the masses of the formula, before its core was added, or of the standard halos and nothing else changed, and none improves the fit there. Self-interacting dark matter, with the isothermal cores of Kaplinghat, Tulin and Yu (2016) and Ren et al. (2019) and a cross-section falling with speed from about 2 cm^2/g in galaxies to 0.1 cm^2/g in clusters, raises the formula's dwarf error from 0.364 to 0.518 dex, its disk error from 0.150 to 0.167 and its error at the cluster centers, before the hot gas was counted, from 0.106 to 0.217; held out on those points, the training disks choose no self-interaction in all 60 partitions, for the formula's halos and for the standard halos alike. Cores in every halo lower the density where dwarfs and cluster centers need it high; cores that later collapse to high density were not modeled. Halo contraction at full strength (Cautun et al., 2020) raises the standard halos' disk error from 0.166 to 0.175 dex, and concentrations set by galaxy size with the simulated relation of Jiang et al. (2019),

applied directly, raise it to 0.25. In these tests the formula stands for cold, collisionless dark matter whose amount is set by how each galaxy formed, and the regularities it encodes need no new force.

The core raises the question again. Three kinds of dark-matter particle also change the inside of a halo, each by a rule that follows the halo and not the gas, and each with one number that must be the same in every halo: fuzzy dark matter, whose solitons follow a core-halo relation (Schive et al., 2014); warm dark matter, which lowers the concentrations of small halos (Schneider et al., 2012); and self-interacting dark matter, with the isothermal cores above. Under a twentieth protocol, hashed before its held-out run, each took the core's place in the formula. Its number was set on all points of the training disks with the core's weights and carried unchanged to every population, and the marks were the core's three (Section 6.4). None meets them. The training disks choose no fuzziness in any of the 60 partitions; on all data the same holds whether the soliton is joined to the halo at 2, 3 or 4 core radii. They choose a warm half-mode mass of 1.1 to 2.0 × 10^9 suns, a thermal relic of 2.0 to 2.4 keV. That lowers the held-out error over all rotation-curve points from 0.218 to 0.210 dex, short of MOND's relation at 0.200 and of the core at 0.193, and it more than doubles the dwarf error, from 0.370 to 0.769; counts of Milky Way satellites exclude relics below 6.5 keV on their own (Nadler et al., 2021). On all points, unlike on the benchmark's, the training disks do choose self-interaction: 1 cm^2/g at low speed in 13 of 15 disk partitions, with the rule's three coefficients held at their all-disk values. It lowers the held-out error on the dropped points by 14 percent (-0.082 to -0.033) and over all points by 9 percent (-0.034 to -0.005), close to the core's 14 and 12 percent, and on eight dwarf irregulars and 26 SPARC galaxies outside the benchmark it scores 0.322 and 0.401 dex against the core's 0.308 and 0.392. But it raises the error on the benchmark's own points by 5.7 percent (+0.003 to

+0.015), on the dwarfs by 25 percent (+0.036 to +0.150) and at the cluster centers by 26 percent (-0.004 to +0.056). Self-interaction makes the cores that the slow centers of gas-rich disks need, and it puts cores in halos that do not want them; the core set by gas and diffuseness appears only where it is needed. Taken alone, the populations would also choose different particles: no fuzziness for the disks, a fuzzy mass of 3 × 10^-20 eV for the dwarfs and of 2 × 10^-22 eV for the early types. On this benchmark the cores follow the gas and not the halo. The three recipes are the published ones for dark matter alone: the soliton ignores the pull of the baryons, and self-interacting cores that later collapse were not modeled. So this is evidence, not proof, that the particle is cold and collisionless and that the baryons make the cores.

Whether the gate's gas stands in for a history of star formation was tested under a twenty-first protocol, hashed before any score was computed. Among the disks, which all still form stars, the amount of gas does not matter. With the gate changed from the gas fraction times diffuseness to diffuseness alone in any system that holds cold gas, and the core's two numbers refitted, every held-out score is level or slightly lower: 0.1530 against 0.1531 dex on the benchmark's points, 0.340 against 0.344 on the dropped points, 0.189 against 0.193 over all points (-0.009 to +0.001), and 64.4 against 64.5 in lensing. In gas-free dwarfs a history gate and the gas gate differ, and there the data lean to history without deciding. For the eight dwarf spheroidals whose dark-matter density at 150 pc was measured by Read, Walker and Steger (2019), the formula's cusps give an error of 0.21 dex, the published history recipe of Read, Agertz and Collins (2016), with those authors' star-formation times and nothing fitted, 0.13, and self-interaction at 1 cm^2/g 0.25. Fornax, which formed stars for 12 Gyr, has a third of the cusp's density, as the history recipe predicts. With eight galaxies the paired range of the difference includes zero (-0.23 to +0.06). On the benchmark's

own points of the 20 dwarf spheroidals with published quenching times (Weisz et al., 2015), where the recipe moves the prediction by 0.15 dex at most, history and cusps are level, 0.268 against 0.261 dex, and self-interaction is worse by 25 percent (+0.012 to +0.121), as on all 75. The same measurements show a limit of the core. In eight dwarf irregulars the formula with its core predicts a density at 150 pc twice the measured one, 0.32 dex too high on average against 0.78 without the core: the core is in the right galaxies and too shallow at the very center. The formula keeps the gas fraction, which every sample provides; a gate built on history needs a history for every gas-free dwarf, and 20 of the 75 have one.

S11. What remains, and the ten forecasts in full (Section 6.6)

Seven things remain. One, in the MaNGA galaxies, lies mostly in the published inputs; the others cannot be repaired with the data in hand without fitting the test itself. Within single disks the curves still depart from the formula by more than their velocity errors, though by less than from MOND's relation and standard halos (Section 6.3), and on the points that the benchmark's cut removes no account reaches the measurement errors (Section 6.4). The outermost lensing bins lie above the formula, next to the radius beyond which missed satellites raise the signal (Brouwer et al., 2021). Extended dwarfs fall below the formula and compact ones above it, and with the orbits in hand tides do not look like the cause: the slope with radius is -0.43 (-1.08 to +0.15) among the 16 Milky Way satellites with pericenters beyond 50 kpc (Pace, Erkal and Li, 2022), -0.56 among the 20 inside, and -0.41 (-1.04 to -0.02) among the 39 dwarfs without orbits, most of them Andromeda's. Dense MaNGA galaxies fall below it and extended ones above. Within the formula nothing tested removes this. Halo contraction moves their error by 0.003 dex, and at full strength, with the lighter stars of a Chabrier

mix in place of the heavier mix of massive spheroids, it leaves the correlation with radius at 0.50. Halos that grow with size at fixed stellar mass lower the error only from 0.116 to 0.105 dex, at an exponent of 3, and leave the trend. Random errors in the half-light radii would give the trend only if they were about 0.08 dex and the kinematic errors far smaller than quoted. Splitting the pull differently, 30 percent less from the stars and twice as much from the halo, removes the trend at 0.090 dex, but that is two numbers fitted to the same 50 galaxies. Rebuilding their published accelerations from one consistent light profile does as well with nothing fitted (below). The CLASH clusters stay 0.07 dex above it on average, because their stated cut understates how they were chosen. The halos of Milky-Way-size spirals come out too heavy. And blind, the formula gives a cluster sample's typical mass, not each cluster's.

Five forecasts follow, written and hashed before any test. First, for isolated lenses chosen with spectroscopic redshifts, as in the Galaxy And Mass Assembly survey that Brouwer et al. compare with, and scored as here, the chi-square per point falls below 1.5 and the outermost bins no longer lie above the formula; a value above 2 with the excess still there counts against it. Second, among dwarfs with dispersions measured to better than 10 percent, pericenters beyond 50 kpc and no tidal features, the residuals' slope with radius at fixed stellar mass lies between -0.2 and +0.2 dex per dex, against -0.5 now; a slope steeper than -0.3 counts against it. Only five dwarfs of the present sample have such dispersions and pericenters, too few to test it. Third, for clusters chosen by a stated threshold in mass, with their hot gas and brightest galaxies measured, the accelerations at 100 to 600 kpc come out within 0.05 dex of the formula on average, with an error below 0.13 dex; an offset beyond 0.10 or an error above 0.16 counts against it. Fourth, a predicted failure: for star-forming central galaxies of 3 to 10 × 10^10 suns, halo masses from lensing or satellite motions lie 0.1 to 0.3 dex below the

formula's, as for the Milky Way and for the blue centrals of Mandelbaum et al. (2016), which the formula overshoots by 0.34 and 0.13 dex at 10^10.56 and 10^10.85 suns. Fifth, with independent, deeper sizes and stellar masses for the 50 MaNGA galaxies, the correlation of their residuals with radius falls below 0.3; if it stays above 0.45, the formula gives the stars too much weight in dense central galaxies. These five and the sixth below were written for the formula before its core was added; the core changes none of the quantities they name by as much as 1 percent.

The fifth forecast has been tested, with the Sérsic fits of the NASA-Sloan Atlas (Blanton et al., 2011) in the MaNGA catalog of the seventeenth SDSS data release (Abdurro'uf et al., 2022). Tian et al.'s half-light radii turn out to be the atlas's elliptical Petrosian radii, to 0.006 dex, and their Sérsic indices the atlas's own, so the atlas's Sérsic fits, from the same images, are the only alternative at hand. Their half-light radii are 0.25 dex larger and their stellar masses 0.09 dex higher. As worded, with only the stellar accelerations rebuilt, the forecast fails: the residuals' correlation with radius is 0.51 (95 percent bootstrap range 0.21 to 0.72), above the 0.45 that counts against it. But Tian et al. derive the measured accelerations from the same light profile, through the Jeans equation, and the forecast left that out. Their construction, the projected Sérsic mass inside the radius of the acceleration and an isotropic Jeans solution with Abel inversion of their fitted dispersion profiles, reproduces their published accelerations within 0.02 dex at one radius, 0.57 to 0.93 times the tabulated last radius; scoring the halo there instead changes the benchmark error by under 0.001 dex. Rebuilt that way from one Sérsic profile throughout, for light, mass and size, the 50 galaxies give the formula an error of 0.087 dex instead of 0.116 (paired bootstrap difference -0.028, 95 percent range -0.053 to -0.008). That is below the 0.112 dex the published errors alone would produce, so those errors overstate the scatter.

The correlation with radius falls to 0.28 (95 percent range -0.04 to 0.55), or 0.35 to 0.37 if the dispersions are held flat beyond one to two half-light radii rather than extended linearly. Standard halos improve alike, to 0.088 dex, and MOND's relation stays at 0.17. Sizes and indices swapped without the matching masses make the error worse, 0.143 dex, so the gain comes from consistency, not from any one input. This full rebuild was chosen after the worded test failed. A sixth forecast, hashed before any test, therefore commits to it on deeper images: rebuilt the same way from single-Sérsic fits of images at least one magnitude deeper than SDSS's, such as those of the DESI Legacy Imaging Surveys, the 50 galaxies give an error below 0.10 dex and a correlation below 0.3; an error above 0.116 or a correlation above 0.45 counts against it. If deeper images agree with the atlas, the published accelerations, not the formula, carry the MaNGA pattern.

A seventh and an eighth forecast concern the gas-rich disks of Section 6.4, the seventh hashed with the looser-halo term's check on the dropped points and the eighth after the nineteenth protocol's run. On at least ten gas-rich dwarf irregular or low-surface-brightness disks in neither SPARC nor the sample of Iorio et al., with resolved HI rotation curves corrected for asymmetric drift, 3.6 micrometer luminosities and sizes, and every point kept, the formula with its core, every coefficient frozen, has a lower error than without it, with a paired bootstrap range that excludes zero, and a lower error than MOND's relation (the eighth forecast); the looser-halo term, which was not adopted, lowers the error as well (the seventh). If the core raises the error, it is wrong; if it lowers it but not below that of MOND's relation, the claim against MOND's relation on such points fails.

A ninth forecast, hashed after the twentieth protocol's run, separates the two readings of the core that survive on gas-rich disks (Section 6.5). A core made by the baryons is absent where there is no gas; a core made by self-interaction is present in every halo. On at least ten gas-poor dwarf galaxies in none of this

paper's samples, the 264 of Section 3.6 among them once their motions are measured, each with a stellar mass, a half-light radius and a dynamical mass at or inside that radius, the formula with every coefficient frozen has a lower error than the same halos with self-interaction of 1 cm^2/g, with a paired bootstrap range that excludes zero. If the self-interacting halos have the lower error, the cores do not follow the gas and the reading of Section 6.5 is wrong.

A tenth forecast, hashed after the twenty-first protocol's runs, is the one the formula is most likely to fail. The formula gives a gas-free dwarf a cusp whatever its history. On at least four gas-free dwarfs outside the eight of Read, Walker and Steger (2019) whose stars formed over at least 9 Gyr, each with a dark-matter density measured inside 0.3 half-light radii, the measured density lies within 0.2 dex of the formula's on average. If it lies lower by 0.2 dex or more, as it does in Fornax, the core follows the history of star formation and the gate must be rebuilt on it, with the recipe of Read, Agertz and Collins (2016) as the first candidate.

S12. Limits, next tests, and the contents of the data and code repository (Section 8 and Data availability)

The comparison has limits. It fits no raw lensing shear, cluster temperature, cosmic microwave background or large-scale structure likelihood; a complete case for either side must, and relativistic MOND theories already compute cosmological spectra (Skordis and Zlosnik, 2021). Nor does it address the S8 tension, the lower clumpiness some weak lensing surveys find than the cosmic microwave background predicts (Pantos and Perivolaropoulos, 2026). Whatever settles it must still produce halos that follow starlight as found here. MOND's external-field effect enters only through the estimate of McGaugh and Milgrom (2013), which takes the host's field as uniform; a full solution along each satellite's orbit is left for later work. Dwarf masses rest on equilibrium estimators. The halos

use a median concentration relation and no formation histories. Each protocol was fixed before its run on data inspected in earlier runs. Section 5.3's cluster-scale test models no cluster selection and does not recover halo masses; Section 6.1's formula models it, but its blind test recovers only typical cluster masses, and CLASH's stated cut understates its clusters' masses by a factor of two. Within those limits the three tests agree.

Brouwer et al. (2021) find that, at the same stellar mass, early types show lensing accelerations 0.17 dex (split by Sérsic index) to 0.27 dex (by color) above those of late types, a difference of 5.7 sigma or more. Its authors note that a gravity modification independent of other galaxy properties could not explain this, and that circumgalactic gas equal to the stellar mass around the early types alone would reduce it to about 4 sigma, since equal stellar mass need not mean equal baryonic mass. The stellar rule's size term gives a quarter to a third of the split (Section 3.8); the rest comes from massive late types whose lensing falls below the rule. Two tests follow: halo mass against size at fixed stellar mass and type, where the rule predicts lighter halos for larger disks, and a halo whose concentration, rather than mass, depends on size, fitted to rotation curves and scored on lensing; the lensing profile, flatter than predicted (Section 3.8), also calls for a freer concentration, and the mix of relaxed and unrelaxed halos in a stack accounts for part of it (Section 6.1). Dwarfs with full Jeans profiles and modeled external fields should keep the stellar rule ahead, or overturn it. A fuller halo model should be scored the same way: the bend in the stellar-to-halo relation above the Milky Way's mass, size measured against its expected value at fixed stellar mass, size-dependent concentration, tidal stripping of satellites, using measured Gaia orbits where they exist, and unresolved binary stars, which inflate the dispersions of the faintest dwarfs (McConnachie and Côté, 2010). A modified gravity law that hopes to cover all scales must pass the mass budget of Section 4

and one constant across disks, dwarfs and clusters.

The long version's statement of what the data and code package holds follows. That package is now the public repository named in the paper's Data availability statement.

The supplement contains the frozen inputs, selection and exclusion tables with provenance and hashes, all coefficients, the prediction tables for every fold, the protocols fixed before each run, independent numerical checks and the scripts that reproduce every number here. The disk fits also rebuild from the public SPARC files with independent code, and separate scripts rerun the dwarf test on a fixed version of the Local Volume Database, run the out-of-sample test, with both versions of its protocol and their hashes, the learning curve, the external-field, subhalo and error tests, and the third protocol's test on dwarfs no rule had scored, with its forecasts for 264 dwarfs and their hash, the fourth protocol's combined score with the forecast ranges, the fifth protocol's cluster-scale test of the Moster relation with its inputs, the sixth protocol's audit of that test, the seventh protocol's comparison with the published lensing masses, with the transcribed table, the eighth protocol's lensing check of galaxy halos, with its transcribed table, and the ninth protocol's forward model of the KiDS lensing profiles, with the release's files, their hashes and the digitized stellar-mass histograms, post-run checks that hold out each of the 85 dwarfs in turn, and the tenth to sixteenth protocols' one-formula fits, held-out runs and checks, with the HeCS-omnibus tables read for the blind test and their hashes, and the scripts that score MOND's relation and standard halos on the same benchmark (Section 6.3), the noise floors and pattern tests of Section 6.3 (Table 9), the three physical alternatives, the twentieth protocol's test of three kinds of dark-matter particle and the twenty-first protocol's test of a gate built on star-formation history, with the tables read from the sources of Read, Walker and Steger (2019) and Weisz et al.

(2015) and their hashes (Section 6.5), the ten forecasts of Section 6.6 with their hashes, the test of the fifth, with the NASA-Sloan Atlas values for the 50 MaNGA galaxies drawn from the DR17 MaNGA catalog (drpall-v3_1_1.fits, SHA-256 9d34371c...) and the rebuild of Tian et al.'s accelerations, and the seventeenth to nineteenth protocols for gas-rich disks, the last of which adds the formula's core, with the tests of Section 6.4 on the points the benchmark drops and on galaxies outside it, the rotation curves of Iorio et al. (2017) as downloaded, and the fluxes and scale lengths read from the sources of Dale et al. (2009) and Read et al. (2017), with their hashes. It also records two coding errors in its own checks, found by review after the runs and fixed; neither changes a number reported here. It credits the SPARC, LITTLE THINGS, Local Volume Legacy, nearby-galaxy, X-COP, CLASH, MaNGA, NASA-Sloan Atlas, KiDS, HeCS-omnibus and stellar-history catalogs on which it rests. This work collects no new observations.

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